Soft-to-Rigid: Shape Continuation as an Alternative Search for Packing Congruent Shapes

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Abstract

Hardening searches for dense packings by starting congruent pieces as their inscribed disks and continuously deforming them into prescribed polygons while compressing the container. Shape evolution is established in packing physics; our contribution is a controlled study of this finite-container optimization path and its failure boundaries. We compare hardening, rigid starts, growth, simulated annealing and perturbation–compression across seven families and 188 instances. Gradient paths have equal step budgets; the two in-house baselines use reference-hardening pair-evaluation budgets, with measured overshoot and separate postprocessing costs. Tuned rigid starts attain the lowest container size most often overall. Hardening reaches particular square packings missed by the other tested searches within the stated run counts. A targeted higher-budget follow-up adds two such best-known successes, while identifying reused runs explicitly. At equal nominal total run budget, a mixed hardening–rigid allocation solves 139 instances versus 138 for doubled rigid search; this does not demonstrate an advantage or equivalence. A preregistered pilot policy on four additional families likewise shows no detected gain. Exploratory ablations and selected replay measurements describe different search behavior without identifying a causal mechanism. Versioned plans, run outputs, saved analyses and selected replays support inspection. These results characterize an alternative search path under the tested tuning and budget design; they establish neither general competitiveness nor a validated rule for allocating a fixed compute budget.

Keywords: geometric packing; shape continuation; stochastic search; computational experiments; reproducibility

1 Introduction

How small a container holds n copies of a given shape? For unit squares in a square the question goes back to Erdős and Graham (1975); catalogues now cover squares, triangles, hexagons and cubes in squares, triangles, circles and cubes (Friedman, n.d.a; The Squares Project, n.d.), and most entries are best known packings found by search rather than proved optimal. Most searches move rigid pieces: simulated annealing (Kirkpatrick et al., 1983), perturbation and compression (Gensane, 2004; Gensane and Ryckelynck, 2005), growing the pieces in size (Lubachevsky and Stillinger, 1990), overlap minimisation in a shrinking container (Umetani et al., 2009; Gardeyn et al., 2025), or global solvers on exact non-overlap models (Berthold et al., 2026).

This paper changes the pieces instead. *Hardening* starts every piece as the disk inscribed in it, compresses the disks under inward pressure on the container, and deforms them into the target shape while the

pressure continues. It is graduated non-convexity (Blake and Zisserman, 1987) applied to geometry: a disk has no orientation, so the pieces are already densely packed when their corners appear and rotation becomes a choice. The shape family it moves along, a polygon dilated by a disk at fixed inradius, is not new: it is the spheropolygon of discrete-element modelling (Alonso-Marroquín, 2008), and Jin et al. (2015) moved dense periodic packings along its 3D analogue in both directions between tetrahedra and spheres. Our contribution is a controlled finite-container comparison of this optimisation path toward a prescribed polygon, with explicit tuning, budget and postprocessing limits (Sections 3, 5 and 8); it is not a claim to have invented shape evolution. On squares it recovers several classical packings from random starts, and the earlier companion project reported a twelve-cube feasible packing (Nakajima, 2026b); that prior reported bound is not a new record of the controlled study here.

Here we ask whether hardening is a search worth running alongside existing ones. We compare three paths through shape space (hardening, growing, and rigid pieces on the same schedule) and two classical baselines on seven families of shapes and containers (115,696 stored run records in 16 search campaigns, including deterministic C16 reuse rather than that many independent trials, plus one tightening-only campaign), with the explicit step and pair-evaluation budget design of Section 3 and comparable grid-search effort. Since a packing search is judged by the best packing it finds, the comparison is by the *lowest* container size each method reaches in a fixed number of runs, after constrained numerical tightening; the procedure being compared is the batch of runs, not a single run (Birattari and Dorigo, 2007). Our findings:

- Hardening is not a better default. Untuned, it looked better than rigid starts; tuned with comparable effort, rigid starts reach the lowest size most often over the seven families (Table 3).
- It is a different search. On squares in a square it reaches the lowest size on 27 of 28 instances and is the only method that low on 5, including Trump's $n = 11$ and Wainwright's $n = 19$ packings, which rigid starts reach in none of our campaigns. In a targeted follow-up campaign at ten times the budget it is the only one of five methods to reach the best known packings for $n = 17$ and $n = 26$; it also does so at $n = 19$ and 29, but those hardening and rigid runs repeat an earlier campaign (Section 6).
- Exploratory ablations show that neither the disk-compressed start nor matched area growth reproduces hardening's pattern (Table 6), consistent with an effect of the gradual path; the pre-registered tests on squares in a square are inconclusive, and hardening's unique packings end with more tilted pieces (Figure 5) without that isolating orientation as the cause.
- The bias has a cost. Hardening is worse on triangles and on four families withheld from tuning and method analysis, apart from the disclosed preregistration geometry smoke tests, and the pilot policy we tested, which chooses the path per instance from 8 runs of each, showed no detected advantage over always starting rigid (Table 7). An earlier indication that it gains more from longer runs did not replicate.

Written campaign plans, amendments, exploratory analyses and post-hoc checks are identified separately, with registration-timing limits in Online Resource 1. Raw outputs and saved analyses are versioned; selected best/hit trajectories have decimated replays. The original event export covers C00–C10, with later analyses also consuming campaign files directly. Online Resource 1 maps the evidence and distinguishes saved-data inspection from rerunning optimization. Section 7 discusses the scope-limited practical implications and describes downstream reuse of the earlier release.

2 Related work

Changing shape during a packing simulation. Hard-particle physics has long changed particle shape during a run. The event-driven molecular dynamics of Donev et al. (2005), which generalises Lubachevsky–Stillinger growth (Lubachevsky and Stillinger, 1990), explicitly allows time-dependent particle shapes.

"Digital alchemy" (van Anders et al., 2015; Geng et al., 2019) treats shape parameters as thermodynamic variables sampled by Monte Carlo shape moves, to design shapes for a target structure. Closest to our parametrisation, Jin et al. (2015) take spherotetrahedra (a tetrahedron dilated by a ball, with a roundness ratio defined relative to a tetrahedron of the same inscribed sphere) and evolve dense periodic packings along that family in both directions: from tetrahedron packings toward spheres, and, in an "inverse evolution", from sphere packings toward tetrahedra. Superdisk and superball families (Jiao et al., 2008, 2009), truncation families (Engel et al., 2012) and rounded cubes in spherical confinement (Wang et al., 2018) interpolate between round and faceted shapes, but packings are computed separately at each shape. Minkowski rounding is standard in discrete-element modelling as spheropolygons (Alonso-Marroquín, 2008; Galindo-Torres et al., 2010), where it smooths contact geometry but is held fixed. Starting round and becoming faceted is therefore not new. What hardening adds is the setting: the same inradius-preserving family used as an optimisation homotopy toward a prescribed polygon inside a finite container, scored by the smallest container reached and compared against tuned rigid searches with ablations of the path. It is an instance of graduated non-convexity and continuation (Blake and Zisserman, 1987; Allgower and Georg, 1990), for which there is theory on when following a path beats a cold start and when it need not (Mobahi and Fisher III, 2015; Hazan et al., 2016).

Disks as a stand-in for polygons. In operations research, inscribed circles appear as relaxations or surrogates: QP-Nest (Jones, 2014) bounds nesting problems by enforcing non-overlap only on inscribed circles, and sparrow (Gardeyn et al., 2025) builds its overlap proxy from inscribed "poles". In facility layout and floorplanning, modules are first modelled as circles and then converted to exact shapes in a second stage (Anjos and Vannelli, 2002; Luo et al., 2008). These use circles as a fixed surrogate or as a one-shot initial solution. We deform each piece continuously from its inscribed disk while the container is compressed, and test separately whether the disk-compressed start or the gradual path is responsible for any effect (campaign C11).

Packing in containers. Phi-functions (Bennell et al., 2010; Romanova et al., 2018) and separating-line programs (Peralta et al., 2018) give exact non-overlap models; our tightening step is a separating-line program of this kind. Overlap minimisation in a shrinking container dominates irregular strip packing (Umetani et al., 2009; Imamichi et al., 2009; Egeblad et al., 2007; Gardeyn et al., 2025), with jostling as a constructive alternative (Dowsland et al., 1998); analytical placers replace overlap with a smooth density penalty of increasing weight (Lu et al., 2015); and gradient descent on poses with an adaptively shrinking container has recently been applied to irregular 3D objects (Gupta and Raman, 2026). Our energy is of this family; what differs is that the pieces' shapes change along the way.

Records. Best known packings of equal pieces come from Friedman's survey and Packing Center (Friedman, 2009, n.d.a), the Squares Project (The Squares Project, n.d.) and Packomania (Specht, n.d.). Record-finding methods include billiards and growth (Lubachevsky and Stillinger, 1990; Lubachevsky and Graham, 1997), perturbation–compression (Gensane, 2004; Gensane and Ryckelynck, 2005), perturbation-based thresholding search (Lai et al., 2023), periodic adaptive-shrinking-cell and divide-and-concur optimisation (Torquato and Jiao, 2009; Kallus et al., 2010), global MINLP solvers (Berthold et al., 2026), and LLM-driven program search (Novikov et al., 2025; Georgiev et al., 2025; Tencent Hunyuan, n.d.). Berthold et al. (2026) formulate and computationally study regular polygons in regular polygons and Platonic solids in Platonic solids; their supplement lists 12 cubes at 2.94281 (Friedman's 1998 value), and Hyra (Tencent Hunyuan, n.d.) later reports 2.93277. Records for squares in a square at larger n ($n \geq 88$) continue to fall to an iterative "shrink-hopping" search with exact certificates (itsnaka, 2026).

Evaluating stochastic searches. A record search is used for the best packing it finds, so we compare the lowest size each method reaches in a fixed number of runs. Birattari and Dorigo (2007) point out that the best of N runs of an algorithm is one sample from a different algorithm, N restarts of the first, and that this algorithm should be judged by its behaviour over repeated batches, not by one batch's best. We therefore treat "32 runs of a method" as the procedure under comparison, report repeated independent batches where we have them, and use resampling only as a sensitivity check, in the spirit of performance profiles and time-to-target plots (Dolan and Moré, 2002; Aiex et al., 2007; Hansen et al., 2021). Lai et al. (2023) report best, mean and success rate over 20 runs against Packomania; we report explicit budget units, tuning allocations and plan-specified decision rules.

3 Method

Pieces and paths. A piece is a regular polygon P with unit side and inradius ρ (a unit square, triangle or hexagon), or a unit-diameter disk. Its *shape state* is

$$P(\gamma,\tau) = \gamma\big[(1-\tau)\,P \;\oplus\; \tau\rho\,D\big],$$

the polygon scaled about its incenter, dilated by a disk of radius $\tau\rho$ (a spheropolygon (Alonso-Marroquín, 2008)), and scaled by γ. At $\tau = 1$ it is the inscribed disk, at $\tau = 0$ the polygon, and for $\gamma \le 1$ every state lies inside the polygon, so moving toward $(\gamma,\tau) = (1,0)$ only grows the piece. Two facts describe what the family does. Its support function is

$$h_{P(\gamma,\tau)}(u) = \gamma\big[(1-\tau)\,h_P(u) + \tau\rho\big],$$

for a unit direction u, so the dependence of a piece's extent on its orientation is scaled down by $(1-\tau)$: rotating a nearly round piece barely changes the energy, and orientation becomes a decision only late in the path. And since $A(P) = \frac{1}{2}\,\text{perimeter}\cdot\rho$, its area at $\gamma = 1$ is

$$A(\tau) = A(P) - \big[A(P) - \pi\rho^2\big]\tau^2,$$

so a hardening piece gains the area outside its inscribed disk, 21% of a square, 9% of a hexagon and 40% of a triangle. A *path* is a schedule for (γ,τ) during the middle phase of a run (Figure 1). We compare three:

- **harden**: $\gamma = 1, \tau : 1 \to 0$ (disks harden into polygons);
- **grow**: $\tau = 0, \gamma : \gamma_0 \to 1$ (rigid polygons grow, as in Lubachevsky–Stillinger (Lubachevsky and Stillinger, 1990));
- **rigid**: $(\gamma,\tau) = (1,0)$ throughout, on the same schedule;

and two classical baselines on rigid pieces: **sa**, Metropolis simulated annealing over poses and container size (Kirkpatrick et al., 1983), and **pc**, perturbation–compression (Gensane, 2004; Gensane and Ryckelynck, 2005) (shrink, relax, perturb on failure). Two further paths serve as ablations: **snap** compresses disks exactly as harden does and then switches to the polygons at once ($\tau : 1 \to 0$ in one step), and **grow-area** keeps rigid polygons whose area follows hardening's area schedule ($\tau = 0$, $\gamma^2 = 1 - (1 - \pi\rho^2/A)\,t^2$ with $t : 1 \to 0$).

Energy. The container (square, equilateral triangle or circle) is centred at the origin with size L: side length for a square/equilateral triangle, radius for a circle. For poses (x_i, y_i, θ_i),

$$E = \sum_{i<j} \big(2r - \mathrm{sd}_{ij}\big)_+^2 + \sum_i \sum_v \big(\text{wall violation of } v\big)_+^2 + \mu L,$$

One gradient-path run. Let B denote the budget multiplier; base compression, morph and settling lengths are $n_C, n_M, n_S = 2500, 7500, 1500$ steps.

1. Initialize random poses in a container of area 3.2 times the pieces' area; set (γ, τ) to the path start.
2. Compress for Bn_C steps with pressure 3μ and decaying noise.
3. Follow the path for Bn_M steps: update (γ, τ) with normalized progress, and take momentum steps at pressure μ with decaying noise.
4. Settle the final pieces for Bn_S steps with no noise, decaying pressure as $\mu 10^{-4s/(Bn_S)}$.
5. Legalise by scaling centres apart until the pair-overlap tolerance is met, then measure the smallest container.

Figure 1: One gradient-path run with the step schedule used in the study

where $r = \gamma\tau\rho$, sd_{ij} is the signed distance between the cores (exact vertex–edge distance when apart, minus the separating-axis penetration depth when overlapping) and v ranges over core vertices. The term μL is inward pressure. Positions, angles and L follow momentum gradient descent with analytic gradients and decaying Gaussian noise. In 3D the same holds for cubes with quaternion orientations and the 15-axis separating test (Nakajima, 2026b).

Step details. Each momentum step updates velocities as $v \leftarrow 0.85\, v - 0.04\, \nabla E$ (container size: $v_L \leftarrow 0.85\, v_L - 0.02\, (\mu + \partial E/\partial L)$), caps a piece's translation step at $0.03\sqrt{2}\, R$ (R the circumradius) and its rotation at 0.03 rad, then adds Gaussian noise. With noise multiplier ν (Appendix, frozen settings), the positional noise is $0.015\, \nu\, (1 - u)$ during compression and $0.004\, \nu\, (1 - u)$ during the path, where $u \in [0, 1]$ is progress through the phase, and the angular noise is 1.5 times the positional noise. The implementation is `engine/engine2d.js`.

Legalisation, tightening and certification scope. Penalties tolerate small overlaps, so every run is *legalised*: centres are scaled apart about their mean until no pair overlaps by more than 10^{-12} under the separating-axis test, and the smallest container (translation free) is measured. Near-best runs are then *tightened* by constrained numerical optimisation: minimise L over poses and one separating line (plane) per nearby pair (Peralta et al., 2018), with every vertex inside the container, by SLSQP (Kraft, 1988) in trust-region outer iterations, keeping a step only if the independently legalised size decreases. Tightening is a local method and gives no optimality guarantee; it only removes the slack a penalty method leaves. The below-catalogue certification route (not applied to every numerically reached catalogue value) spreads a tightened packing below a catalogue value to a clearance of 10^{-6} and checks it twice: in floating point, and in exact rational arithmetic. The exact check uses rational rotations from $\tan(\theta/2)$, replaces each piece by a rational polygon scaled by $1 + 10^{-8}$ and proves, in $\mathbb{Q}(\sqrt{3})$, that this polygon contains the true unit polygon with margin 10^{-9}, and exhibits a rational separating direction for every pair. The current exact 2D checker supports circles, triangles, squares and hexagons; it does not certify the pentagon/octagon families in C15. A numeric reached classification is not an exact certificate or an optimality proof. No below-catalogue claim folder was produced by this controlled study. Online Resource 1 checks four separately enlarged square witnesses; those certificates do not establish feasibility at their original contact objectives.

Budgets. A run's budget is its number of steps: every gradient path takes $11{,}500\, B$ steps. Pair evaluations then differ by path, since soft pieces have more near neighbours: at $n = 20$ squares in a square a hardening run makes about $1.2\times$ the pair evaluations of a rigid run (Table 11 in the appendix), which, if anything, favours hardening. The two baselines are stopped once their pair evaluations exceed those of a reference hardening run at the same n (one fixed seed, with the baseline's own pressure and noise settings); for squares

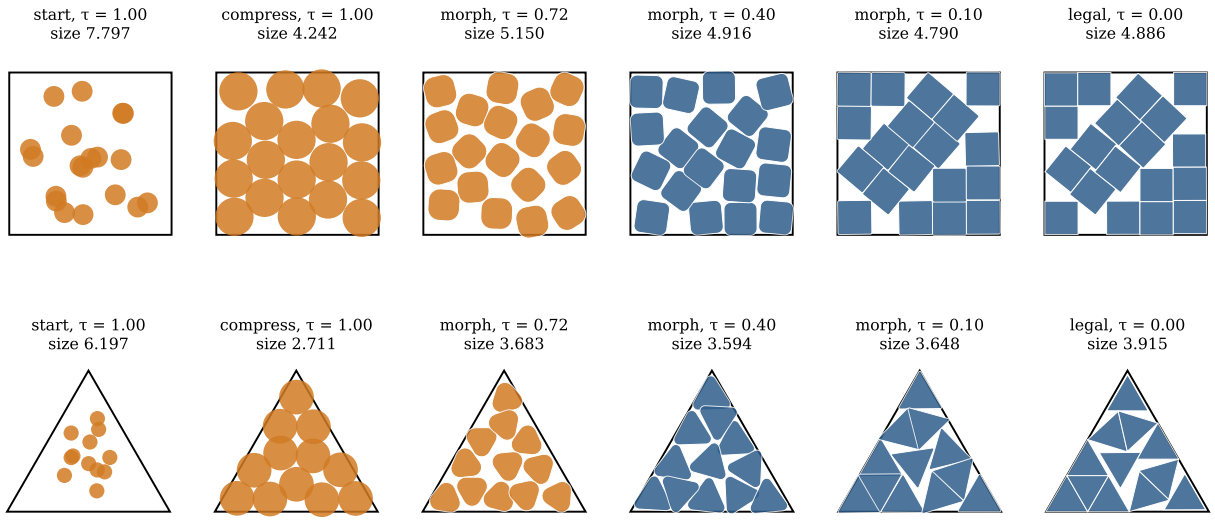


Figure 2: Two hardening runs (C07). Top: 19 squares; disks are compressed, harden into squares, and settle into Wainwright's packing (side $3 + 4\sqrt{2}/3$), which no other method reached in 32 runs. Bottom: the best of 32 hardening runs for 12 triangles in a triangle ends at 3.915, while rigid and snap starts reach the best known 3.879; the triangles that emerge from the near-hexagonal disk arrangement keep orientations that do not tile. These are selected decimated replays from versioned campaign files. The triangle size 3.915 is the legalised simulator endpoint before tightening.

in a square this reference is 438,032 pair evaluations at $n = 20$ and 696,566 at $n = 30$, 1.05 and 1.07 times the mean of the 32 tuned hardening runs. Simulated annealing adds a final short gradient relaxation; perturbation–compression stops at the end of a relaxation. For the square-in-square diagnostics in Table 11, mean pair evaluations divided by the saved reference allowance are 1.02476 and 1.02520 for annealing at $n = 20, 30$, respectively, and 1.57588 and 1.68891 for perturbation–compression. These are means for those two cells, not global maxima. In recorded simulator wall-clock time on one core of the study machine (Node.js; tightening excluded), hardening was the fastest of the five methods in the two square-in-square diagnostic cells, $n = 20$ and 30 (Table 11). This timing observation is specific to the measured machine and simulator workload. It excludes tuning, repeated reference runs, postprocessing and verification, so it does not establish equal end-to-end cost or general wall-clock superiority.

Endpoint and assistance definitions. For non-truncated values, a run reaches the catalogue when its selected size is no more than the value plus 10^{-9}. For truncated values, the ledger verdict uses a gap strictly below the stored printed-precision tolerance (default 10^{-5}), while the later saved statistical/policy code uses size no greater than value plus tolerance. Lowest-size ties use relative 10^{-9}, with an additional 10^{-12} absolute allowance in the C07 descriptive count code; C15 strict-lower comparisons use $x < y(1-10^{-9})$. These numeric rules do not certify equality or optimality. Substantive AI assistance in implementation, analysis, writing and preparation review is described in Statements and Declarations.

4 From squares to cubes

The method was developed on unit squares in a square, where records are dense and audited (The Squares Project, n.d.; Friedman, 2009). With the original schedule, hardening reaches Trump's packing for $n = 11$

		original settings			2D-tuned settings		
n	reference	harden	grow	rigid	harden	grow	rigid
9	2.70711	**2.70711**	**2.70711**	**2.70711**	**2.70711**	**2.70711**	**2.70711**
10	2.70711	2.76777	2.76777	2.89443	**2.70711**	2.76777	**2.70711**
11	2.88295	**2.89443**	2.91739	2.91899	2.91250	2.91328	2.94281
12	2.93277	2.94301	2.98587	3.00000	**2.94281**	2.94751	3.00000
13	2.95600	2.99497	3.00000	3.00000	**2.95615**	3.00000	3.00000
14	2.98995	**3.00000**	**3.00000**	**3.00000**	**3.00000**	**3.00000**	**3.00000**

Table 1: Cubes in a cube (C05): lowest side over 32 runs at the prior search budget, after tightening. The reference column excludes the prior reported $n = 12$ feasible bound (2.93152), from a separate 192-start search; this is not a current-study catalogue target. Bold: lowest in the row.

(Trump, 1979) after tightening (to within 10^{-14}) in 4 of 80 random starts at three times the base budget and 2 of 80 at the base budget; rigid starts with the same schedule reach it in none of 80 at either budget (campaign C00).

An earlier note by the author, *Twelve unit cubes in a cube of side 2.93152* (unpublished note), reported a feasible packing of 12 unit cubes in a cube of side 2.9315185. The public code and certificate are separately available (Nakajima, 2026b). Its public repository compares this bound with the catalogued Hyra value 2.9327718 and Friedman's 1998 value 2.94281 (Tencent Hunyuan, n.d.; Friedman, n.d.b); these are historical comparator snapshots, not a current priority adjudication. The author's public build log flags novelty relative to Haowei Lin's arrangement as disputed (Nakajima, 2026a). We therefore use the earlier bound only as context and make no new-record or undisputed-priority claim; its priority is unnecessary to the controlled comparison here. Campaign C05 repeats the cube experiment with three gradient paths and two settings at the earlier bound's search budget (Table 1). Hardening has the lowest side of the three paths at $n = 11$, 12 and 13 under both settings, all paths reach Friedman's $n = 9$ packing (Friedman, n.d.b), and with tuned settings hardening and rigid starts also reach his $n = 10$ packing $2+1/\sqrt{2}$, which the original schedule never did. Rigid starts end at the trivial side 3 for $n \geq 12$. In these 32 seeds nothing went below a catalogue value: the best side at $n = 12$ is 2.942809 (grow 2.947506, rigid 3.000000), while the earlier feasible bound was reported from a separate 192-start search. The two searches have different run counts and configurations, so this non-reproduction alone does not estimate a discovery probability. At $n = 13$ hardening reaches 2.956145, consistent with Friedman's 2.956+ (Friedman, n.d.b) to the precision printed.

Two results made the broader question worth asking. Rigid starts in these first experiments used the hardening schedule untouched, so they were never given a fair chance. And squares in a square may be special: many of its best packings tilt a few squares against their neighbours, exactly the configurations a late commitment to orientation should help with.

5 A wider test

Design. Seven families: unit squares in a square, a circle and a triangle; unit triangles in a triangle and a square; unit hexagons in a square; and, as a control, unit-diameter disks in a square, where hardening changes nothing because a disk is already a disk. Best known values come from the Squares Project (The Squares Project, n.d.), Friedman's catalogue (Friedman, n.d.a) and Packomania (Specht, n.d.); truncated catalogue values ("2.956+") count as reached within their printed precision. All n from 2 to 30 (to 20 for hexagons) are used except five reserved for tuning, 188 instances in all.

Each method was tuned twice, first by median gap on held-out instances (C01–C03; a 3×3 grid, extended past its edge by a fixed rule, so harden, sa and pc saw 27 settings and grow and rigid 18), then by the lowest

treatment	harden	grow	rigid	sa	pc
no tightening	6	19	3	1	0
C07 policy (within 2%, up to 3 runs)	24	20	18	14	2
equal (best 3 runs of every method)	23	20	19	17	4

Table 2: Instances on which each method's lowest size is the lowest of all five, on the 28 instances of C14, under three tightening treatments of the same C07 runs.

size reached (C06; one 3×3 grid per method, 16 runs per setting on each of eight held-out instances of squares in a square and in a circle and triangles in a square and in a triangle, scored by the mean over instances of the lowest of the 16 runs). The second round used larger instances ($n > 30$) than the test set, so its settings may be less suited to small n; Table 4 breaks the results down by n. Each method then ran 32 times per instance at the nominal allocation described in Section 3, and in each instance and method the best run (if within 2% of the best known value) and any run within $2 \cdot 10^{-3}$ of it, at most three, were tightened. Plans recorded intended comparisons, with the registration-timing limits and amendments documented in Online Resource 1; the disclosed C15 geometry smoke tests preceded its plan. Changes after a campaign started are recorded as amendments.

Result. Table 3 counts, per family, the instances on which each method's lowest size is the lowest of all five. Rigid starts reach the lowest size most often overall (155 instances, against 144 for grow and 142 for harden) and reach the best known value most often (133 against 129 and 125). As a check of sensitivity to the particular seeds, we resampled each instance's 32 runs with replacement (1000 times) and recounted. Within each resample, rigid's count exceeds hardening's in 99% of resamples (difference 3 to 24, 95% range) and grow's in 99%, while grow against harden goes either way (-13 to 9). This measures sensitivity to the seed pool on these instances, not uncertainty about future campaigns: a resample cannot contain a run better than the ones observed, it omits a cell's single best run with probability $(31/32)^{32} \approx 0.36$, which is why every method's resampled count is lower than its observed one, and tightened sizes are fixed by the original pool rather than re-selected in each resample. Over instances solved by exactly one of the two, harden against rigid is 6 vs 14 (exact sign test, $p = 0.12$). C04 and C07 specified one-sided tests in favor of hardening. For C07 against rigid starts the registered value is $p_1 = 0.979305$, so that directional hypothesis is not supported; the descriptive two-sided value is $p_2 = 0.115318$. Later C11–C13, C15 and C16 plans specify two-sided tests. With the doubled-minimum-tail convention, $p_2 \leq 2p_1$ for either one-sided direction; reversing the observed direction makes the registered p_1 large, not this inequality false. Online Resource 1 reports the omitted C04/C07 grow, annealing and perturbation–compression comparisons and the remaining plan-specified tests. Simulated annealing improved most from tuning for the lowest point; perturbation–compression stays far behind at this budget, partly because its best run was within reach of tightening on only 56% of instances (against 97% to 100% for the gradient paths; Table 11). The similar eligibility fractions for rigid and hardening limit one obvious source of asymmetry, but alone do not establish that tightening cannot affect their relative ordering. To check whether it decides the rest of the comparison, campaign C14 tightened the three best runs of every method on the 28 instances fixed in advance for C13 ($n = 11, 17, 23, 29$ per family; hexagons $n = 8, 12, 16, 20$), with no eligibility condition. Under equal tightening the baselines gain a few instances and the order on these 28 instances, where hardening happens to lead, is unchanged (Table 2); the subset is too small to rule out larger changes for sa and pc elsewhere. Without any tightening the order is different, so comparisons of penalty methods on legalised sizes alone would mislead.

No run went below a published value, and 43 instances were solved by no method. The median-tuned comparison (C04, Table 10) also puts rigid starts first.

family	N	**harden**	**grow**	**rigid**	**sa**	**pc**
squares in square	28	**27 (5)**	21	23 (1)	21	17
squares in circle	28	16 (6)	**17 (4)**	15 (3)	14 (2)	8
squares in triangle	29	28	27	**29**	24	8
triangles in triangle	28	18	24 (2)	**26 (4)**	19	10
triangles in square	28	14 (2)	17 (3)	**22 (7)**	9	5
hexagons in square	18	15	14	**16 (1)**	14 (2)	9
circles in square (control)	29	**24**	**24**	**24**	**24 (2)**	15 (1)
all	188	142 (13)	144 (9)	**155 (16)**	125 (6)	72 (1)
reached best known		125	129	133	115	71

Table 3: Lowest point by family (C07; every method tuned for its lowest point, 32 runs per instance, stated step/evaluation allocations). Each cell: number of instances on which the method's lowest tightened size is the lowest of all five methods; in brackets, on how many it is the only method that low. Bold: most per family.

n	instances	harden	grow	rigid	sa	pc
2–10	58	54	54	56	56	47
11–20	70	50	57	52	43	20
21–30	60	38	33	47	26	5

Table 4: Instances on which each method's lowest size is the lowest of all five, by n (C07). Small instances are mostly ties; differences open up from $n \approx 11$. Rigid starts lead most at $n = 21$–30, nearest the $n > 30$ instances used for lowest-point tuning, so part of that lead may be a tuning-domain effect; the median-tuned comparison (C04, Table 10) also puts rigid starts first.

How often does a method get there? A count of instances hides how close each call was. Figure 3 shows every instance: most instances are ties; squares in a square is the only family where every instance but one that differs favours hardening, triangles lean towards rigid starts, and squares in a circle alternate. Figure 4 shows, for six instances, the share of k-run subsets of the observed 32 runs that include a run reaching the best known value. Hardening reaches Trump's $n = 11$ and Wainwright's $n = 19$ packings in 2 and 3 of 32 runs, and rigid starts in none; on triangle instances the picture is reversed. At $k = 32$ the curve is 1 whenever any run succeeded, which says nothing about a fresh batch. Repeated batches do: on the 28 instances of C13, harden and rigid each have three independent 32-run batches at the base budget, and hardening reaches Trump's $n = 11$ packing in 2 of 3 batches and Schadt's $n = 29$ packing in 2 of 3, rigid starts in 0 and 0; on the triangles-in-triangle instance $n = 29$ it is the other way round (hardening 1 of 3, rigid 3). Over all 28 instances, the best known value is reached in 49 of the 3×28 hardening batches and 46 of the rigid ones, and on 4 (harden) and 2 (rigid) instances some batches reach it and others do not. Statements that one method is "the only one" to reach a packing are statements about one batch of 32 runs.

Inference scope. The unit of the paired summaries is an instance in a designed benchmark, not an independently sampled application. Families share geometry and the experiments include selected follow-ups. Multiple family/endpoint comparisons and exploratory analyses are therefore interpreted descriptively; nominal sign-test values do not provide family-wide error control. A best-of-32 result is one realization of a restart procedure, and seed-pool resampling cannot establish reliability on new instances or independent batches. Ties and catalogue tolerances are defined by the saved analysis code. No nonsignificant test is treated as evidence of equivalence.

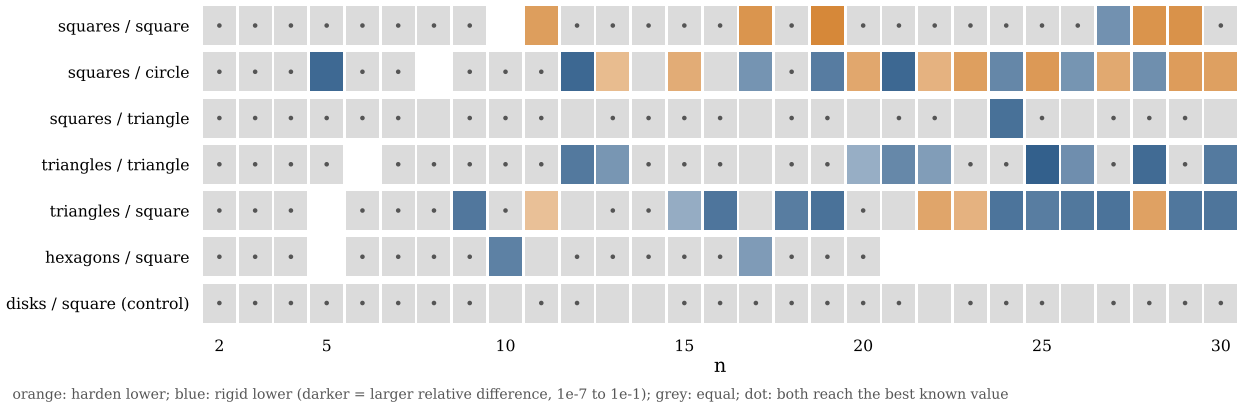


Figure 3: Hardening against rigid starts on every instance (C07): which of the two reaches the lower size, and by how much. Blank cells were held out for tuning.

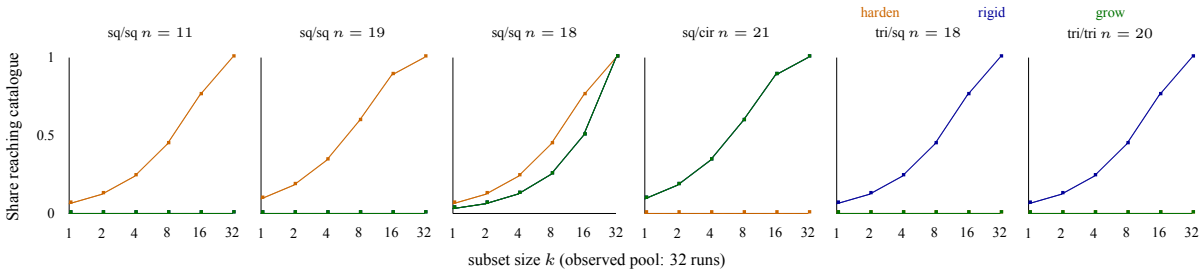


Figure 4: Share of k-run subsets of the observed 32 runs (C07) that include a run reaching the best known value, $1 - \binom{32-s}{k} / \binom{32}{k}$ for s successful runs, on six instances where the gradient paths differ. This is a without-replacement estimate from one pool; it is 1 at $k = 32$ whenever $s > 0$. Runs not selected for tightening count with their legalised size.

6 What we learned

Where hardening differs. For squares in a square, hardening's lowest size is the lowest of all methods on 27 of 28 instances and the only one that low on 5; with identical settings, rigid starts manage 23. In 32 runs each, hardening alone reaches Trump's $n = 11$ and Wainwright's $n = 19$ packings (Figure 2). For triangles the order reverses: rigid starts are lowest on 22 instances of triangles in a square against 14 for hardening, and 26 against 18 in a triangle (Figure 2). On the control family the gradient paths reach the lowest size on the same number of instances, as they should.

With more time on squares. Campaign C16, a targeted follow-up planned after C07 and C09, tested at ten times the base budget the advantage already seen on squares in a square ($n = 17$–30, 16 runs per method; Table 5). Hardening's lowest size is the lowest of the five on 13 of 14 instances (annealing 9, rigid 8, grow 7, pc 6), and it reaches the best known value on 12 (annealing 9, rigid 8). Within this 16-run-per-method C16 comparison, hardening is the only method to reach the best known packing for $n = 17, 19, 26, 29$; annealing is the only one for $n = 27$. Tightening the best run of every method at the four hardening-only instances, regardless of the tightening gate, leaves every other method short of the best known value (`scripts/c16_equal_tighten.py`). Against rigid starts hardening is lower on 6 instances and higher on 0 ($p = 0.031$), but the hardening and rigid runs at $n = 19$, 28 and 29 repeat C09's exactly (same seeds and

n	best known	harden	grow	rigid	sa	pc
17	4.67553	Y	0.324	0.032	0.032	0.092
18	4.82288	Y	0.177	Y	Y	0.015
19†	4.88562	Y	0.114	0.114	0.114	0.114
20	5.00000	Y	Y	Y	Y	Y
21	5.00000	Y	Y	Y	Y	Y
22	5.00000	Y	Y	Y	Y	Y
23	5.00000	Y	Y	Y	Y	Y
24	5.00000	Y	Y	Y	Y	Y
25	5.00000	Y	Y	Y	Y	0.756
26	5.62132	Y	0.379	0.379	0.086	0.202
27	5.70711	0.098	0.293	0.139	Y	0.294
28†	5.82444	**0.042**	0.176	0.073	0.178	0.179
29†	5.93383	Y	0.066	0.066	0.066	0.066
30	6.00000	Y	Y	Y	Y	Y

Table 5: Squares in a square at ten times the base budget (C16, 16 runs per method). Y: reaches the catalogue value within tolerance; otherwise the gap to it. Bold: lowest of the five. †: hardening and rigid runs repeat C09 exactly (same seeds and budget, deterministic runs).

budget, deterministic runs), and all three are hardening wins; on the remaining 11 instances the count is 3 against 0 ($p = 0.25$). C16 adds new successes against all four other methods at $n = 17$ and $n = 26$ rather than six independent confirmations.

Separating the disk start from the gradual path. Hardening differs from rigid starts in two ways: pieces are compressed as disks, and they then harden gradually. Campaign C11 separates the two with *snap*, which compresses disks exactly as hardening does and then switches to the polygons in one step, with the same seeds, settings and budget (Table 6). Its pre-registered test compares, per family, the instances that exactly one of harden and snap solves. On squares in a square only two instances are discordant, so the test is inconclusive ($p = 0.5$). The plan's decision rule then attributed hardening's advantage there to the disk-compressed start. That rule was wrong: failing to detect a difference does not show that there is none, still less which mechanism is responsible. We record the rule and its output, and draw no causal conclusion from it.

Exploratory comparisons of the sizes reached, which were not the pre-registered test, are as follows. We detect no difference between snap and rigid starts (each lower on 20 and 20 of 188 instances). Hardening and snap differ in opposite directions by family: counting instances where hardening is lower against those where snap is lower, squares in a square give 6–0, triangles in a triangle 0–8 and triangles in a square 6–14 (all families, 17 against 36, $p = 0.013$). The pooled exploratory lowest-size comparison favors snap (17 hardening-lower versus 36 snap-lower, $p = 0.0126603$), opposite to the square-in-square direction (6 versus 0). These are family-dependent descriptive differences; they do not identify a causal orientation mechanism. A confirmatory directional test on squares would need more discordant instances than the registered reached-endpoint comparison provides.

Campaign C12 asks the same of area. A hardening piece also grows, by up to 40% of its area; *grow-area* keeps rigid polygons but scales them so their area follows hardening's schedule exactly. Again we detect no difference from rigid starts (lower on 15 and 14 instances), and hardening differs from it in the same directions as from snap: 5–1 on squares in a square and 1–8 on triangles in a triangle (all families 20 against 31, $p = 0.16$). Its registered reached-endpoint test on squares in a square has 2 hardening-only versus 0 grow-area-only successes ($p = 0.5$), again inconclusive; the 5 versus 1 lowest-size comparison

family	N	harden : snap	snap : rigid	harden : grow-area	grow-area : rigid
squares in square	28	6 : 0	1 : 2	5 : 1	2 : 1
squares in circle	28	5 : 11	10 : 6	9 : 8	8 : 4
squares in triangle	29	0 : 1	0 : 0	0 : 1	0 : 0
triangles in triangle	28	0 : 8	0 : 2	1 : 8	0 : 3
triangles in square	28	6 : 14	8 : 9	5 : 11	5 : 6
hexagons in square	18	0 : 2	1 : 1	0 : 2	0 : 0
disks in square	29	0 : 0	0 : 0	0 : 0	0 : 0
all	188	17 : 36	20 : 20	20 : 31	15 : 14

Table 6: Which part of hardening matters (C11, C12; harden and rigid from C07, same seeds, settings and budget). Each cell $a : b$ counts the instances on which the first method's lowest size is lower than the second's, and the reverse; the rest are ties. *snap* compresses disks as hardening does, then switches to polygons at once; *grow-area* keeps rigid polygons whose area follows hardening's.

($p = 0.21875$) is exploratory. Neither the disk start nor the area schedule alone reproduces hardening's pattern of results.

What the pieces do. Figure 5 measures three things on the replay of each instance's best run (lowest legalised size, before tightening), for hardening and rigid starts: net displacement of pieces between the end of compression and the end of the run, net rotation over the same interval, and the share of pieces that end tilted against every container edge. Midway through the morph phase, where a hardening piece has $\tau \approx 0.5$, the orientational order of squares in a square, $|\langle e^{4i\theta} \rangle|$, has a median of 0.53; at the same step of a rigid run it is 0.81, so rigid squares are more aligned at that point than hardening squares. On the 5 squares-in-square instances where hardening alone reaches the lowest size, its final packing has more tilted squares than rigid's best on the same instance every time (5 of 5; median shares 0.45 against 0.28), as the classical packings at $n = 11$, 17 and 19 require. On triangles in a triangle hardening pieces move about twice as far after compression as rigid ones (median 0.53 against 0.28 unit sides), and on the 4 instances where rigid starts alone reach the lowest size the gap is the same (0.54 against 0.30). These are descriptive. They fit two hypotheses: that the support function's factor $1 - \tau$ lets hardening defer the choice of orientation until neighbours are in place, which squares need when a block must tilt against aligned neighbours; and that triangles, whose corners stick out twice as far as their inradius ($R/\rho = 2$, against $\sqrt{2}$ for a square) and whose good packings alternate up and down, pay for that deferral with large rearrangements. The area explanation alone (a triangle's inscribed disk covers 60% of it, a square's 79%) is weakened by hexagons, whose disk covers 91% and where hardening is no better than rigid starts (15 against 16 of 18, mostly ties). Changing the path also changes contact geometry and forces, so none of this isolates orientation as the cause.

Does choosing the path transfer to new families? If the better path depends on the family, a search could choose it per instance. Campaign C15 tests that as a decision on four families not used for tuning or method analysis: pentagons and octagons in a square, hexagons in a triangle and triangles in a circle, $n = 6$–20 (60 instances). Before the plan was written, one geometry smoke-test run at $n = 8$, seed 3, was made for each family and path; this exception qualifies the held-out description. Each instance then gets 64 hardening and 64 rigid runs, from which three pre-registered policies of 64 runs are formed: always rigid, always harden, and a pilot that spends 8 runs on each path and the remaining 48 on the path whose pilot reached the lower legalised (untightened) size (Table 7). The pilot and always-rigid rarely differ (lower on 2 and 3 instances, $p = 1$), so by the plan's rule we report no detected benefit from choosing the path. Always-harden is worse than always-rigid (3 against 17, $p = 0.0026$), most clearly on pentagons, and reaches the best known value on

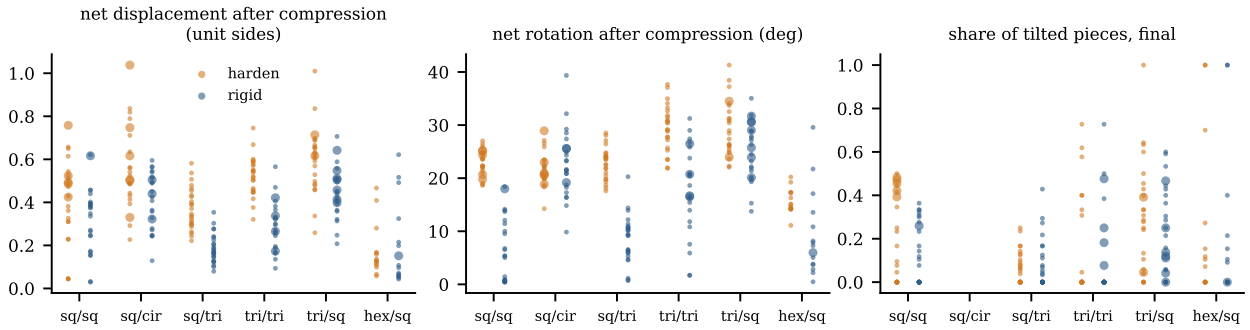


Figure 5: What pieces do after compression, on the best run of each C07 instance with $n \geq 6$ (orange: hardening; blue: rigid; large dots: the method alone reached the lowest size). Left: mean net displacement; middle: mean net rotation, modulo the piece's symmetry (for hardening this includes turning while pieces are still nearly round and orientation costs almost nothing); right: share of pieces with no edge within 2° of a container edge (square and triangle containers).

family	N	harden : rigid	pilot : rigid	pilot chose harden	reached (r / h / p)
pentagons in square	15	1 : 8	0 : 1	3	8 / 4 / 8
octagons in square	15	0 : 3	0 : 0	6	11 / 9 / 11
hexagons in triangle	15	0 : 2	0 : 1	8	10 / 9 / 10
triangles in circle	15	2 : 4	2 : 1	3	11 / 10 / 12
all	60	3 : 17	2 : 3	20	40 / 32 / 41

Table 7: Held-out families (C15), $n = 6$–20, 64 runs per policy. $a : b$ counts instances where the first policy's result is lower than the second's and the reverse. The pilot spends 8 runs on each path, then 48 on the one whose pilot went lower. Reached (r / h / p): instances where always-rigid, always-harden and the pilot reach the best known value.

32 instances against 40 for rigid starts and 41 for the pilot. The pilot picked hardening on 20 instances and so avoided most of its losses, but was lower than always-rigid on only 2 instances. The pilot compares untightened sizes, which Table 2 shows can misrank methods; in a post-hoc check that compares the tightened best pilot run of each path instead, the pilot picks hardening on 3 instances rather than 20, changes 17 decisions, and still differs from always-rigid on 2 against 3 instances (`scripts/c15_tightened_pilot.py`). None of these families shows a family-level advantage for hardening (it is lower on 3 individual instances), so the test says that this pilot policy found no gain on these families; it is not a test of whether a selector can find a favourable family, since none was present.

Partly soft starts. Varying only the starting softness τ_0 on the 40 most contested instances (Table 8), the counts of instances where each value reaches the lowest size are 7, 15, 18 and 17 for $\tau_0 = 1, 0.5, 0.25, 0$. Squares in a square favour soft starts (3, 4, 2, 0); triangles favour rigid or slightly rounded ones (2, 7, 8, 10). These instances were selected where hardening and rigid starts disagreed, mostly in rigid's favour, which tilts this count toward small τ_0; partly rounded starts are a candidate default, not an established one.

Longer runs, or more runs? An earlier test (C09, appendix) suggested that hardening gains more than rigid starts from longer schedules, but it changed run length and restart count together and used instances chosen because the two paths disagreed. Campaign C13 repeats it cleanly on 28 instances fixed in advance ($n = 11, 17, 23, 29$ per family; hexagons $n = 8, 12, 16, 20$), with 32 runs at three times the budget, and 96

family	instances	$\tau_0 = 1$ (harden)	0.5	0.25	0 (rigid)
squares in square	6	3	4	2	0
squares in circle	13	2	3	6	5
squares in triangle	1	0	1	1	1
triangles in triangle	7	0	4	3	5
triangles in square	12	2	3	5	5
hexagons in square	1	0	0	1	1
all	40	7	15	18	17

Table 8: How soft the start must be (C08): on the 40 most contested instances of C07, the number of instances where each starting softness τ_0 reaches the lowest size (32 runs each, all other settings equal; ties counted for every tied value).

runs at the base budget for the same nominal total step allocation. Counting instances where hardening is lower against those where rigid starts are lower: 7 against 2 with 32 base runs ($p = 0.18$), 4 against 3 with 32 long runs, and 6 against 2 with 96 base runs. Longer runs improved rigid starts more than hardening (rigid lower than its base on 5 instances and higher on 0; hardening 4 and 3), and at equal nominal step allocation, three times longer runs and three times as many runs did about equally well for both paths (3 against 3 for hardening, 3 against 2 for rigid). The C09 result does not replicate; we withdraw it.

Two paths, or more runs of one? The paths fail on different families, so running both might help. At equal total budget, 32 rigid runs plus 32 hardening runs solve 139 of the 188 instances, against 138 for 64 rigid runs and 137 for 32 rigid plus 32 growing runs (instances solved by exactly one of the first two: 4 vs 3, two-sided $p = 1$). On squares in a square alone the mixed portfolio solves one instance more than doubled rigid runs: it gains $n = 11$ and 19 and loses $n = 27$. The C10 plan specified a one-sided sign test; its report gives $p = 0.5$, whereas the two-sided descriptive value reported here is $p = 1$. We detected no benefit from this allocation. These data establish neither an advantage nor equivalence. Equal total budget here means nominal run/step allocation, not equal complete CPU time including tuning and tightening.

Tuning decides comparisons. Our first experiments compared hardening with rigid starts that inherited the hardening schedule, on two square instances, and hardening looked better. Tuned with comparable effort, rigid starts led overall. Re-tuning for the lowest point instead of the median changed four of the five methods' settings (stronger pressure for the gradient paths, more noise for rigid starts) and helped simulated annealing most. It kept the broad pattern, hardening for squares in a square and rigid or growing starts for triangles, though the leading method changed for squares in a circle and triangles in a square.

7 When to use it, and downstream use

When to add hardening. Under these in-house implementations, tested settings and budgets, rigid starts remain the better default. Hardening is a candidate additional path on squares in a square, where it reached packings none of the other four methods reached (Section 6) and where the methods cover different instances: at ten times the budget, hardening and annealing together reach every best known value from $n = 17$ to 30 except $n = 28$. Our hypothesis is that it helps where good packings tilt a block of pieces against an otherwise aligned arrangement, but that is not a validated predictor: we have no test that identifies favourable families in advance, and we have not shown that splitting a fixed budget between hardening and rigid starts beats spending it on rigid starts (C10). The tested settings favor rigid starts on triangles and the additional families; this evidence does not support replacing them with hardening as a default.

Downstream use. The earlier cube release (Nakajima, 2026b) has been reused by others; we report this separately from our results, which it does not change. A continuation of the unfinished $n = 13$ cube search (@BuilderOfAgents, 2026) builds on our simulator, reuses our tightening code and checkers unchanged, adds two moves (re-melting a finished packing: soften, expand slightly, reharden; and inserting a thirteenth cube into the 12-cube packing), and after 2,627 runs reports a certified packing of side 2.956149 at 10^{-6} clearance, consistent with Friedman's $2.956+$ and not claimed as an improvement; C05 reaches a comparable numerical contact value (2.956145) without clearance. Zarzuelo (2026) generalises the construction to all five Platonic solids as pieces and containers, with exact feasibility checks over $\mathbb{Q}(\sqrt{2}, \sqrt{5})$ and an archive search that repeatedly softens and rehardens existing packings; the reported values are certified upper bounds, and comparisons with Berthold et al. (2026) must convert between edge and circumradius normalisations. Yudhiono (2026) reimplements ball-to-hypercube continuation independently in four dimensions and combines it with structured seeds, constrained polishing and basin hopping, reporting floating-point candidates for 26–30 tesseracts. Together they continue a partially explored search, extend the geometry and the exact verification to new solids, and reimplement the method in a higher dimension. None is a controlled comparison, and the first two inherit our simulator (one modified, one generalised) and tightening scheme, so they are not independent validations of it; the generalised exact checker is new verification work. What they add is a use we did not test: softening as a rearrangement operator inside a larger search rather than only as a cold start. Whether temporary softening escapes a rigid local optimum better than rigid perturbation at equal budget, from shared starting packings, is the experiment they motivate.

Other settings. The same question arises wherever an overlap penalty is minimised in a shrinking container: irregular nesting (Wäscher et al., 2007; Bennell and Oliveira, 2008; Leao et al., 2020), where inscribed circles are already an overlap proxy (Jones, 2014; Gardeyn et al., 2025); floorplanning, where circles-then-rectangles is an established two-stage switch (Anjos and Vannelli, 2002; Luo et al., 2008); and faceted particles in confinement (Torquato, 2018; Wang et al., 2018). In each, C11 is the test to run first, and real objectives add stability, orientation and ordering constraints (Bortfeldt and Wäscher, 2013; Araújo et al., 2019) that any gain in density must survive.

8 Limitations and open directions

Limitations. These are search results, not optimality proofs, and tightening is a local method. In the main comparison (C07) each instance had 32 runs per method at a fixed budget, and C16 had 16; "only one method reached it" holds within the run count and budget stated for that campaign. Gradient paths are matched in steps, not pair evaluations or wall-clock time (hardening was the fastest per run in the two square-in-square diagnostic cells); the baselines are matched to a hardening run's pair evaluations, which perturbation–compression overshoots (Table 11). Simulated annealing and perturbation–compression are our implementations; their best runs were less often close enough to be tightened, so the tightening policy favours the gradient paths, most for pc (C14 shows the order survives equal tightening on 28 instances); in C16 the other methods' best runs at the hardening-only instances were tightened after the fact and remained short of the best known value. The lowest-point tuning used instances larger than the test set, and in that round all five winners sat at corners of their two-axis grids and the grid was not extended. Comparable grid effort is not proof of comparable tuning adequacy; simulated annealing moved furthest between the two tuning rounds, so its results are best read as a lower bound on what a wider grid would give. Its step size adapts every 200 moves to keep the acceptance rate between 20% and 40%, so at $T_0 = 10^{-3}$ it behaves as a Metropolis walk with small adaptive steps rather than one with large uphill moves. The grids' ranges were first chosen during development on squares in a square, and settings suited to triangles could lie outside them; the C08 sweep of starting softness is only a partial check. In 3D only the three gradient paths were compared.

Catalogue values come from captions and witnesses as of October 2026; truncated values are only known to their printed precision. Not detecting a difference (between mixed and single-path portfolios, between snap or grow-area and rigid starts, or between the pilot and always-rigid) is not evidence of equivalence at these sample sizes.

Open directions.

- *A rule that predicts the path.* The pilot showed no detected benefit on held-out families that had no aggregate hardening advantage. A useful rule would have to predict, before running, the families where it does; we have one such family and no rule that singles it out in advance.
- *External baselines.* Our simulated annealing and perturbation–compression are our own implementations. Published alternatives require a compatibility audit before a fair comparison. Berthold et al. (2026) provide regular-polygon/container and cube variants that can be compared after explicit scale conversion; their Section 4 variable-radius, maximum-sum circle objective does not match our congruent-disk control; their Section 5 unit-circle/variable-ellipse model is a different container problem. Sparrow (Gardeyn et al., 2025) minimizes strip length at fixed width rather than the finite container scale studied here; adapting that objective would be a changed experiment. No external competitiveness claim is made without such a matched test.
- *Softening as a move.* Re-softening an existing packing and rehardening it, as downstream searches do (@BuilderOfAgents, 2026; Zarzuelo, 2026), against rigid perturbation from the same packings at equal budget.
- *Other paths.* Paths through other shape families (an ellipse for a rectangle, a rounded octahedron for a cube, superellipses (Jiao et al., 2008)), paths that harden pieces one at a time, or schedules that adapt τ to the current overlap.
- *Larger n, other solids, other solvers.* Most unsolved instances have $n \geq 17$. Our controlled study did not include solids other than cubes; downstream work now explores them (Zarzuelo, 2026) and higher dimensions (Yudhiono, 2026), without controlled baselines. Hardening could also seed exact models (Berthold et al., 2026) or shrink-and-separate nesters (Gardeyn et al., 2025).
- *Non-convex and mixed pieces.* The rounded-shape construction needs a convex core; non-convex pieces would need a different family of intermediate shapes, for example a convex decomposition hardened part by part.

Statements and Declarations

Data and code availability. Study code, campaign plans, raw run outputs, saved analyses and selected replays are available at `https://github.com/yoheinakajima/soft-to-rigid/tree/07e3cbd3c4855e5e19344885fbf672c3d7a0f039`. The original event export covers C00–C10; later analyses also read versioned campaign files directly. Online Resource 1 describes the evidence map, limitations of certification and a saved-output verification route. A new optimizer campaign is not needed to inspect the reported stored results. Code is distributed under the repository's MIT license; third-party records and witnesses retain their respective rights. Saved-output verification artifacts are pinned at `https://github.com/yoheinakajima/soft-to-rigid/tree/9d3b33242558eb8d36f6508b394cb016df2cd26b/submission/joh-20261102/verification`; `scripts/verify_joh_packet.py` and `scripts/analyze_joh_reviews.py` at that commit provide its saved-output and review-arithmetic routes. These files are distinct from the original study baseline.

Funding and role of the funder. This research, including computational resources and model/API usage, was supported by Untapped Capital. The author is its founder and general partner and directed the research and AI-assisted workflow; the supporter was therefore not independent of the author.

Competing interests. Yohei Nakajima has a financial interest through his founder and general partner role at Untapped Capital, the disclosed supporter. He created ActiveGraph, used to organize portions of the study evidence. Untapped Capital publicly lists Explore Science as a portfolio company; the study's revision history records feedback from Explore Science. These research, software and investment relationships are disclosed; no independent endorsement or peer review is implied.

Author contributions. Yohei Nakajima is the sole human author. The study and manuscript were developed under his direction with the AI assistance described below. Responsibility and accountability for the work remain with him. The computational checks described here were performed by software in an AI-assisted workflow.

Ethics and consent. This study reports computational geometry experiments and does not involve human or animal participants; participant-consent and human/animal ethics-approval declarations are not applicable to the reported work.

AI assistance and review. Claude (Anthropic) assisted development of the original study's code, experiments, verification, analyses, manuscript and arXiv packaging/PDF preparation. Historical commits identify the tool as "Claude Opus 5.5" and include Claude Code session trailers; this is a recorded label, not independently authenticated provider telemetry. Exact API identifiers and runtime versions are unavailable. Revision history records Explore Science feedback, but the underlying review models and complete original review payloads are unverified. November journal preparation used Codex (OpenAI), including a `gpt-6.1-sol` worker configured with high reasoning effort and manager AI review, for writing/revision, code and evidence audits, verification-program development, compilation and rendered-page checks. The exact manager model is not recorded here. Two separately prompted preparation critiques used Claude (Opus 5.5 Extra UI label) and Gemini (Pro UI label); exact backend revisions are unverified. Their source-grounded dispositions and scope limits are recorded in the verification package. Experimental packing outcomes are produced by the numerical engines. The saved-input verifier and its review were AI-assisted; separate floating-point and exact software checks are not independent human verification or journal peer review. AI tools are not authors. Online Resource 1 and the packet's tool-provenance record distinguish the assistance, observed checks and unresolved version details.

Related work and overlap. Public v1 is `https://arxiv.org/abs/2610.14463v1`, distributed under CC BY 4.0. The earlier twelve-cube note is acknowledged in the text; its public code and certificate record are separately cited. Its theorem, coordinate table, cube figure and exact-certificate derivation are not reproduced here. This article distinguishes that earlier reported feasible bound, including its unresolved public priority attribution, from the present controlled study.

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A Campaign and diagnostic details

Campaigns. C00 reproduce the earlier square results; C01–C03 median tuning; C04 breadth (median-tuned); C05 cubes; C06 lowest-point tuning; C07 breadth (lowest-point-tuned, main); C08 starting softness; C09 longer schedules (confounded); C10 portfolio; C11 snap; C12 grow-area; C13 clean budget study; C14 equal tightening; C15 held-out families (path choice); C16 squares in a square at ten times the budget. Plans, amendments, raw runs, reports and the decision recorded for each are in `campaigns/` and the journal.

Frozen settings. Median-tuned (C01–C03): harden $\mu = 0.08$, noise $\times 8$; grow $\mu = 0.04$, $\gamma_0 = 0.2$; rigid $\mu = 0.04$, noise $\times 2$; sa $T_0 = 3 \cdot 10^{-3}$, $\mu = 0.0025$; pc kick 0.05, relax 19200 steps. Lowest-point-tuned (C06): harden and rigid $\mu = 0.16$, noise $\times 8$; grow $\mu = 0.08$, $\gamma_0 = 0.1$; sa $T_0 = 10^{-3}$, $\mu = 0.02$; pc kick 0.05, relax 19200 steps. All are in `methods/*.json`, each with the campaign and grid point that selected it.

Longer schedules, first attempt (C09). C09 gave hardening and rigid starts three and ten times the budget on the 20 instances where the two differed most. It changed run length and restart count together (32 runs at budget 1, 16 at budgets 3 and 10) and selected instances for disagreement, mostly in rigid's favour, so part of its change is regression toward an even split. We report it (Table 9) but draw conclusions about run length only from C13, which fixes these confounds.

Figure 6 shows per-instance lowest sizes relative to the catalogue.

budget	harden lowest	rigid lowest	harden reaches best known	rigid reaches best known
1×	3	17	1	10
3×	5	16	2	10
10×	9	16	6	9

Table 9: Longer schedules (C09) on the 20 most contested instances: number of instances where each path reaches the lower of the two lowest sizes (ties counted for both), and where it reaches the best known value. Budget 1 is C07 (32 runs); budgets 3 and 10 use 16 runs.

family	N	**harden**	**grow**	**rigid**	**sa**	**pc**
squares in square	28	**28 (7)**	21	20	10	17
squares in circle	28	16 (7)	11 (2)	**19 (9)**	10	8
squares in triangle	29	27	24	**29 (1)**	13	7
triangles in triangle	28	19 (1)	22 (3)	**24 (4)**	8	10
triangles in square	28	10 (1)	**19 (5)**	18 (7)	12 (1)	5
hexagons in square	18	15	15 (1)	**17 (2)**	8	6
circles in square (control)	29	27	27	**28**	11 (1)	11
all	188	142 (16)	139 (11)	**155 (23)**	72 (2)	64 (0)
reached best known		121	119	132	69	63

Table 10: Settings tuned by median gap (C04). Same layout as Table 3.

method	steps (sa: moves) per run		pair evaluations per run		ms per run		best run
	$n = 20$	$n = 30$	$n = 20$	$n = 30$	$n = 20$	$n = 30$	tightened
harden	11,500	11,500	418,365	650,532	451	699	97%
grow	11,500	11,500	426,807	668,192	727	1,183	98%
rigid	11,500	11,500	348,285	550,494	633	1,001	100%
sa	66,854	104,043	448,878	714,120	582	983	94%
pc	19,148	20,530	690,285	1,176,434	1,117	1,993	56%

Table 11: Work per run (C07, squares in a square, mean over 32 runs), wall-clock milliseconds per run on one core (Node.js), and the share of instances on which each method's best run was within 2% of the best known value and so was tightened (all families). Gradient paths are matched in steps; the baselines stop once they exceed a reference hardening run's pair evaluations.

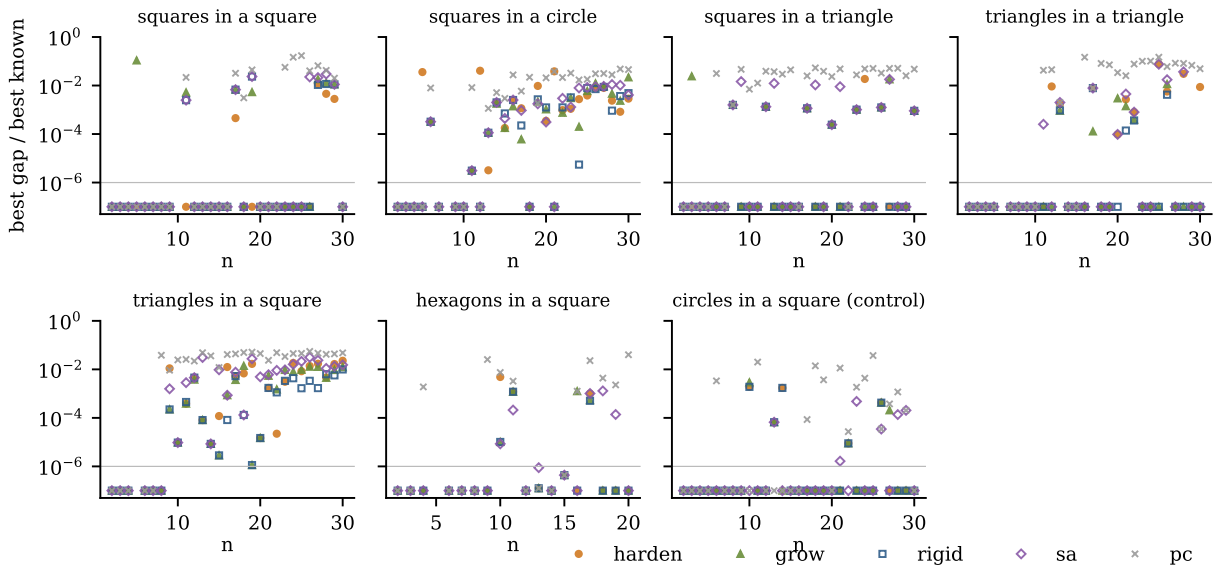


Figure 6: Lowest size reached per instance and method, relative to the best known value (C07). Points on the bottom row reach it.